



四川大學
SICHUAN UNIVERSITY

海納百川 有容乃大

Mathematical Programming

An Introduction to Optimization (Part 2)

Changxi Wang

Sichuan University-Pittsburgh Institute
Chengdu, China

2026–2027 Academic Year



Theorem on Bounded Linear Functions

Theorem

If linear function $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$ is bounded from below/above (i.e., $f(\mathbf{x}) \geq \mu$) over the convex polyhedron $F = \{\mathbf{x} : \mathbf{Ax} \leq \mathbf{b}\}$, then $f(\mathbf{x})$ obtains a minimum/maximum over F , which occurs at one of the extreme points of F

Proof: If $\mathbf{x} \in F$, then $\mathbf{x} = \sum_{i=1}^d \alpha_i \mathbf{x}_i + \sum_{i=1}^{\ell} \beta_i \mathbf{y}_i$, where $\alpha_i \geq 0$ $\beta_i \geq 0$ $\sum_{i=1}^d \alpha_i = 1$. Let $f(\mathbf{x}_k) = \min\{f(\mathbf{x}_1), \dots, f(\mathbf{x}_d)\}$

$$f(\mathbf{x}) = f\left(\sum_{i=1}^d \alpha_i \mathbf{x}_i + \sum_{i=1}^{\ell} \beta_i \mathbf{y}_i\right) = \sum_{i=1}^d \alpha_i f(\mathbf{x}_i) + \sum_{i=1}^{\ell} \beta_i f(\mathbf{y}_i) \geq \sum_{i=1}^d \alpha_i f(\mathbf{x}_i) \geq \sum_{i=1}^d \alpha_i f(\mathbf{x}_k) = f(\mathbf{x}_k)$$

Quiz: Minimize Linear Function

- Minimize $f(x_1, x_2) = -x_1 + x_2$, subject to the constraints

$$-x_1 + 2x_2 \leq 10$$

$$5x_1 + 3x_2 \leq 41$$

$$2x_1 - 3x_2 \leq 8$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Quiz: Matrix Notation

- Minimize $f(x) = c^T x$, subject to $Ax \leq b$, $x \geq 0$

Note that $\mathbf{x} = [x_1, x_2]^T$, $\mathbf{c} = [-1, 1]^T$, $\mathbf{b} = [10, 41, 8]^T$
[

LP Problem: Slack Variables

Finding the optimal solution to the LPP $\rightarrow$ Finding extreme points of the convex polyhedron of F

- Change inequality constraints to equality constraints via slack variables

$$a_{1,1}x_1 + \dots + a_{1,n}x_n + x_{n+1} = b_1$$

$$a_{2,1}x_1 + \dots + a_{2,n}x_n + x_{n+2} = b_2$$

$$\vdots$$

$$a_{m,1}x_1 + \dots + a_{m,n}x_n + x_{n+m} = b_m$$

where $x_i \geq 0, \forall i \in \{1, 2, \dots, n + m\}$



LP Problem: Standard Form

1. $\min \hat{c}^T \hat{x}$, $\hat{A}\hat{x} = \mathbf{b}$, $\hat{x} \geq 0$, where $\hat{c} = [c_1, \dots, c_n, 0, \dots, 0]^T$
2. Feasible set: $\hat{F} = \{\hat{x} \in \mathbf{R}_{(n+m) \times 1} : \hat{A}\hat{x} = \mathbf{b}, \hat{x} \geq 0\}$
3. Coefficient matrix

$$\hat{A} = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} & 1 & 0 & \dots & 0 \\ a_{2,1} & \dots & a_{2,n} & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & \dots & a_{m,n} & 0 & 0 & \dots & 1 \end{bmatrix} = [\mathbf{A} | \mathbf{I}_m]$$

Mapping between F and $\hat{F}$

- Define a function $\sigma : F \rightarrow \hat{F}$, i.e.,

$$\sigma(\mathbf{x}) = \hat{\mathbf{x}} = \begin{bmatrix} \mathbf{x} \\ \mathbf{b} - \mathbf{A}\mathbf{x} \end{bmatrix}$$

- Proposition 8: $\sigma(\cdot)$ is a one-to-one mapping that preserves convex combinations of vectors, i.e.,

$$\sigma(\alpha x + (1 - \alpha)y) = \alpha \sigma(x) + (1 - \alpha) \sigma(y)$$

- Proposition 9: $\mathbf{x}$ is an extreme point of $F \iff \hat{\mathbf{x}}$ is an extreme point of $\hat{F}$



Example: Mapping

$$F = \{[x_1, x_2]^T : x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}$$

$$\hat{F} = \{[x_1, x_2, x_3]^T : x_1 + x_2 + x_3 = 1, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0\}$$

$$\sigma([1, 0]^T) = [1, 0, 0]^T, \quad \sigma([0, 1]^T) = [0, 1, 0]^T, \quad \sigma([0, 0]^T) = [0, 0, 1]^T$$



Theorem on Extreme Points of F

Theorem

If $\hat{x}$ is an extreme point of $\hat{F}$ and $\Delta = \{j : |\hat{x}_j| > 0\}$, then the columns $\{\hat{A}^{(j)} : j \in \Delta\}$ is linearly independent

If

- (a) $\{\hat{A}^{(i)} : i \in \Delta\}$ is some collection of m linearly independent columns of $\hat{A}$
- (b) B is the matrix whose columns are $\{\hat{A}^{(i)} : i \in \Delta\}$
- (c) x^* is a solution to $Bx = b, x \geq 0$

Then the vector $\hat{x}$ whose i -th entry

$$[\hat{x}]_i = \begin{cases} 0, & i \notin \Delta \\ x^*_i, & i \in \Delta \end{cases}$$

is an extreme point of $\hat{F}$



Example: Reformulation with Slack

- Minimize $f(x_1, x_2) = -x_1 + x_2$, subject to the constraints

$$-x_1 + 2x_2 \leq 10$$

$$5x_1 + 3x_2 \leq 41$$

$$2x_1 - 3x_2 \leq 8$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

- Reformulation: $\min\{-x_1 + x_2 + 0 \cdot x_3 + 0 \cdot x_4 + 0 \cdot x_5\}$, subject to $x_i \geq 0, \forall i \in \{1, \dots, 5\}$, and

$$\begin{bmatrix} -1 & 2 & 1 & 0 & 0 \\ 5 & 3 & 0 & 1 & 0 \\ 2 & -3 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 10 \\ 41 \\ 8 \end{bmatrix}$$

Example: Finding Extreme Points

At most $\binom{5}{3} = 10$ sets of three linearly independent columns in the coefficient matrix
Select the first three columns ($x_4 = x_5 = 0$)

$$\mathbf{B} = [\hat{\mathbf{A}}^{(1)}, \hat{\mathbf{A}}^{(2)}, \hat{\mathbf{A}}^{(3)}]$$

$x_1 = 7, x_2 = 2, x_3 = 13$ all nonnegative $\mathbf{x}^* = [7, 2, 13]^T \rightarrow \hat{\mathbf{x}} = [7, 2, 13, 0, 0]^T$ is an extreme point of $\hat{F} \rightarrow [7, 2]^T$ is an extreme point of F



Example: Invalid Basis

- Select $\hat{A}^{(1)}, \hat{A}^{(2)}, \hat{A}^{(4)}$ ($x_3 = x_5 = 0$)

$$\mathbf{B} = [\hat{\mathbf{A}}^{(1)}, \hat{\mathbf{A}}^{(2)}, \hat{\mathbf{A}}^{(4)}] = \begin{bmatrix} -1 & 2 & 0 \\ 5 & 3 & 1 \\ 2 & -3 & 0 \end{bmatrix}$$

Solving $\mathbf{B}\mathbf{x} = \mathbf{b}$ gives $x_1 = 46$, $x_2 = 28$, $x_4 = -273$ (negative), so $[46, 28, 0, -273, 0]^T \notin \hat{F}$, hence $[46, 28]^T$ is NOT an extreme point of F .



Primal Simplex: Equality Form

Any LLP can be expressed as:

$$\min \mathbf{c}^T \mathbf{x} \text{ subject to } \mathbf{Ax} = \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0}$$

$\mathbf{A} \in \mathbf{R}_{m \times n}$ $\mathbf{x} \in \mathbf{R}_{n \times 1}$, and $\text{rank}(\mathbf{A}) = m \leq n$

All the constraints are equality constraints

Each $\mathbf{x}_i$ is restricted to be nonnegative

Equivalent problem with $z_o = \mathbf{c}^T \mathbf{x}$: Finding a solution $\mathbf{x} \in \mathbf{R}_{n \times 1}$ and $z_o \in \mathcal{R}$

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ -\mathbf{c}^T \mathbf{x} + z_o &= 0 \end{aligned}$$

for $\mathbf{x} \geq \mathbf{0}$ and z_o is minimal



Primal Simplex: Matrix Equation

Matrix equation

Assume $\text{rank}(\mathbf{A}) = m$:

$$\mathbf{A} = \underbrace{[\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(m)}, \mathbf{A}^{(m+1)}, \dots, \mathbf{A}^{(n)}]}_{:=\mathbf{B}} = \mathbf{D}$$

Initial matrix for the problem:

$$\left[\begin{array}{c|c} & A, b, c, z_o \end{array} \right] \left[\begin{array}{c|c} & A, b, c, z_o \end{array} \right] \left[\begin{array}{c|c} & A, b, c, z_o \end{array} \right] \left[\begin{array}{c|c} & A, b, c, z_o \end{array} \right]$$



Primal Simplex: Initial Tableau

- Left multiplying the initial matrix by:

$$\begin{bmatrix} & \end{bmatrix}$$

- It results simplex tableau:

$$\begin{bmatrix} & \end{bmatrix}$$

- Can be alternatively done by a sequence of elementary row operations



Primal Simplex: Simplex Tableau

Above augmented matrix represents the linear system:

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{B}^{-1}\mathbf{D} & \mathbf{0} \\ \mathbf{0}^T & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_D \\ z_o \end{bmatrix} = \begin{bmatrix} \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{bmatrix}$$

$$\begin{cases} \mathbf{x}_B + \mathbf{B}^{-1}\mathbf{D}\mathbf{x}_D = \mathbf{B}^{-1}\mathbf{b} \\ (\mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T)\mathbf{x}_D + z_o = \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{cases} \quad \text{If } \mathbf{B}^{-1}\mathbf{b} \geq 0, \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T \leq 0^T:$$
$$\begin{cases} z_o = \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \\ \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{x}_D = \mathbf{0} \end{cases}$$



Example: Primal Simplex

Question:

$$\min -2x_1 - x_2 + 2x_3$$

$$\begin{array}{rcl} & x_1 + x_3 & = 4 \\ & -2x_1 + x_2 & = 8 \\ \text{subject to} & x_1 & \geq 0 \\ & x_2 & \geq 0 \\ & x_3 & \geq 0 \end{array}$$



Example: Basis Selection

Solution: By elementary row operations,

$$[\quad]$$

In above, $\mathbf{B} = [\mathbf{A}^{(3)}, \mathbf{A}^{(2)}]$



Theorem 1: Optimality Condition

Theorem 1: Solution for $\min \mathbf{c}^T \mathbf{x}$ subject to $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$
If $\mathbf{B}^{-1}\mathbf{b} \geq \mathbf{0}$ and $\mathbf{c}_B \mathbf{B}^{-1} \mathbf{D} - \mathbf{c}_D^T \leq \mathbf{0}^T$, then the feasible solution

$$\mathbf{x}_o = \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_D \end{bmatrix} := \begin{bmatrix} \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{0} \end{bmatrix}$$

is optimal, and $z_o = \min \mathbf{c}^T \mathbf{x} = \mathbf{c}^T \mathbf{x}_o = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b}$

Proof: Suppose an **arbitrary** feasible solution $\mathbf{A}\bar{\mathbf{x}} = \mathbf{b}$ and $\bar{\mathbf{x}} \geq \mathbf{0}$

$$\begin{aligned} \mathbf{b} &= \mathbf{A}\bar{\mathbf{x}} = [\mathbf{B}, \mathbf{D}]\bar{\mathbf{x}} = \mathbf{B}[\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]\bar{\mathbf{x}} = \mathbf{B}\mathbf{x}_B \\ &\Rightarrow \mathbf{B}[\mathbf{x}_B - [\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]\bar{\mathbf{x}}] = \mathbf{0} \\ &\Rightarrow \mathbf{x}_B = [\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]\bar{\mathbf{x}} \\ &\Rightarrow \mathbf{c}_B^T \mathbf{x}_B = [\mathbf{c}_B^T, \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{D}] \bar{\mathbf{x}} \\ &\leq [\mathbf{c}_B^T, \mathbf{c}_D^T] \bar{\mathbf{x}} = \mathbf{c}^T \bar{\mathbf{x}} \end{aligned}$$

$\mathbf{c}_B^T \mathbf{x}_B$ is the minimal value for the problem, and $\mathbf{x}_o$ is optimal.

Basic Solutions and BFS

- Basic solution of $\mathbf{Ax} = \mathbf{b}$:

$$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}, \quad \mathbf{x}_D = \mathbf{0}$$

- Basis matrix: the nonsingular matrix $\mathbf{B}$
- Admissible basis: $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(m)}$
- Basic feasible solution (BFS): a basic solution that is feasible
- Nondegenerate BFS: A BFS for which $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} > \mathbf{0}$
- Theorem 1 gives a sufficient condition, i.e., $\mathbf{c}_B\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T \leq \mathbf{0}^T$, for a BFS $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} > \mathbf{0}$, and $\mathbf{x}_D = \mathbf{0}^T$ to be an optimal solution.



Feasible, Basic, and BFS

- Feasible Solution: "Qualified", wide range candidates

$$\hat{A}\hat{x} = \mathbf{b}, \quad \hat{x} \geq 0$$

- Basic Solution: "Simple & Efficient", not necessarily feasible

$$\hat{A}\hat{x} = \mathbf{b}, \quad \mathbf{x}_D = \mathbf{0}$$

- BFS: "Key candidates" for optimal

$$\hat{A}\hat{x} = \mathbf{b}, \quad \mathbf{x}_D = \mathbf{0}, \quad \hat{x} \geq 0$$



Example: LP with Slack

Question: $\min 10x_1 + 18x_2$ subject to

$$5x_1 + 2x_2 \leq 170$$

$$2x_1 + 3x_2 \leq 100$$

$$x_1 + 5x_2 \leq 150$$

$$x_i \geq 0, \forall i \in \{1, 2\}$$

Reformation:

$$5x_1 + 2x_2 + x_3 = 170$$

$$2x_1 + 3x_2 + x_4 = 100$$

$$x_1 + 5x_2 + x_5 = 150$$

$$x_i \geq 0, \forall i \in \{1, 2, \dots, 5\}$$

Example: Geometric Interpretation

$$\begin{aligned} [& \quad] \quad (10): O, A, B, E, F, G, H, I \\ (5): & O, A, B, C, E \\ (1): & C \end{aligned}$$

Simplex Tableau Details

The Primal Simplex Procedure

Simplex Tableau:

$$\begin{bmatrix} \mathbf{I}_m & \mathbf{B}^{-1}\mathbf{D} & 0 & \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{0}^T & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T & 1 & \mathbf{c}_B^T\mathbf{B}^{-1}\mathbf{b} \end{bmatrix}$$

$$X_B \triangleq [\mathbf{I}_m, \mathbf{B}^{-1}\mathbf{D}]$$

$$X_B = \begin{bmatrix} 1 & 0 & \cdots & 0 & x_{1,m+1} & \cdots & x_{1,j} & \cdots & x_{1,n} \\ 0 & 1 & \cdots & 0 & x_{2,m+1} & \cdots & x_{2,j} & \cdots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & x_{m,m+1} & \cdots & x_{m,j} & \cdots & x_{m,n} \end{bmatrix}$$

$$z_j \triangleq \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{A}^{(j)}, c_j \triangleq [\mathbf{c}]_j, \forall j = 1, \dots, n$$

$$z_j - c_j = \begin{cases} 0 & j = 1, \dots, m \\ [\mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{D} - \mathbf{c}_D^T]_j & j = m + 1, \dots, n \end{cases}$$

Theorem 2: Unboundedness

Theorem 1: If $z_j - c_j \leq 0, \forall j = 1, \dots, n$, the BFS $\{x_B = \mathbf{B}^{-1}\mathbf{b} \geq 0, \mathbf{x}_D = 0\}$ is an optimal solution

What if $z_j - c_j > 0$, for some j ?

↓

Theorem 2: If there exists a j so that $z_j - c_j > 0$ and each $x_{i,j} \leq 0$, then the LLP

$$\min \mathbf{c}^T \mathbf{x} \text{ subject to } \mathbf{Ax} = \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0}$$

has an unbounded minimum over F



Theorem 3 & 4: Pivoting and Optimality

Theorem 3

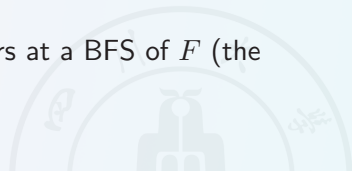
If there exists a j so that $z_j - c_j > 0$, and if some $x_{i,j} > 0$, then by pivoting at the element $x_{k,j}$, a new basic feasible solution is generated that does not increase the value of z_o . Note that k is selected such that

$$\theta \triangleq \frac{[\mathbf{B}^{-1}\mathbf{b}]_k}{x_{k,j}} = \min \left\{ \frac{[\mathbf{B}^{-1}\mathbf{b}]_i}{x_{i,j}} : x_{i,j} > 0 \right\}$$

If $\mathbf{B}^{-1}\mathbf{b} > 0$, i.e., the BFS $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}$ and $\mathbf{x}_D = \mathbf{0}$ is nondegenerate, then the value of z_o decreases.

Theorem 4

If $\mathbf{c}^T \mathbf{x}$ obtains a finite minimum over F , then the minimum occurs at a BFS of F (the minimum may occur at other places as well)



Example: Introducing Slack

Question

$$\min \mathbf{c}^T \mathbf{x}$$

subject to

$$3x_1 + x_3 - x_4 = 1$$

$$7x_1 + x_2 - x_4 = 0$$

$$-8x_1 + x_4 \leq 1$$

$$x_i \geq 0, \forall i$$



Example: Augmented Form

- Add a slack variable

$$\min \mathbf{c}^T \mathbf{x}$$

subject to

$$\begin{bmatrix} & & & & \end{bmatrix}$$
$$\mathbf{c} = [1, -1, 2, -3, 0]^T$$

Example: Basis and BFS

Example 14

$$\begin{bmatrix} 3 & 0 & 1 & -1 & 0 & 0 & 1 \\ 7 & 1 & 0 & -1 & 0 & 0 & 0 \\ -8 & 0 & 0 & -1 & 1 & 0 & 1 \\ -1 & 1 & -2 & -3 & -1 & 0 & 1 \end{bmatrix}$$

Row operation: $\mathbf{A}_{(4)} - (\mathbf{A}_{(2)} - 2\mathbf{A}_{(1)})$

$$\begin{bmatrix} 3 & 0 & 1 & -1 & 0 & 0 & 1 \\ 7 & 1 & 0 & -1 & 0 & 0 & 0 \\ -8 & 0 & 0 & -1 & 1 & 0 & 1 \\ -2 & 0 & 0 & -2 & -1 & 1 & 2 \end{bmatrix}$$

$\mathbf{B} = [\mathbf{A}^{(3)}, \mathbf{A}^{(2)}, \mathbf{A}^{(5)}]$ is nonsingular

$\mathbf{D} = [\mathbf{A}^{(1)}, \mathbf{A}^{(4)}]$

BFS: $[x_3, x_2, x_5]^T = [1, 0, 1]^T, [x_1, x_4]^T = [0, 0]^T$ (degenerate)

Example: Pivot and Unboundedness

Example 14

$$\begin{bmatrix} 3 & 0 & 1 & -1 & 0 & 0 & 1 \\ 7 & 1 & 0 & -1 & 0 & 0 & 0 \\ -8 & 0 & 0 & -1 & 0 & 1 & 1 \\ -2 & 0 & 0 & -2 & 0 & 1 & 2 \end{bmatrix}$$

Row operations: $\mathbf{A}_{(1)} + \mathbf{A}_{(3)}$, $\mathbf{A}_{(2)} + \mathbf{A}_{(3)}$, $\mathbf{A}_{(4)} - 2\mathbf{A}_{(3)}$

$$\begin{bmatrix} -5 & 0 & 1 & 0 & 1 & 0 & 2 \\ -1 & 1 & 0 & 0 & 1 & 0 & 1 \\ -8 & 0 & 0 & -1 & 0 & 1 & 1 \\ -14 & 0 & 0 & -2 & 0 & 1 & 0 \end{bmatrix}$$

(1) Pivot in the 4th column, since $z_4 - c_4 = 2 > 0$

(2) $\min \left\{ \frac{|\mathbf{B}^{-1}\mathbf{b}_i|}{x_{i,4}} : x_{i,4} > 0 \right\} = 1 \Rightarrow k = 3 \Rightarrow \text{pivoting } x_{3,4}$

(3) $\mathbf{B} = [\mathbf{A}^{(3)}, \mathbf{A}^{(2)}, \mathbf{A}^{(4)}]$ and $\mathbf{D} = [\mathbf{A}^{(1)}, \mathbf{A}^{(5)}]$

(4) BFS: $x_3 = 2$, $x_2 = 1$, $x_4 = 1$, and $x_1 = x_5 = 0$



Example: Maximization to Minimization

Question

$$\max 2x_1 + x_2 + 3x_3$$

subject to

$$x_1 + 3x_2 + 2x_3 \leq 6$$

$$-x_1 + x_2 - 3x_3 \geq -9$$

$$x_i \geq 0, \quad \forall i$$

Transformation

$$\min -2x_1 - x_2 - 3x_3$$

subject to

$$x_1 + 3x_2 + 2x_3 + x_4 = 6$$

$$x_1 - x_2 + 3x_3 + x_5 = 9$$

$$x_i \geq 0, \quad \forall i$$



Example: Simplex Iterations

Example 15

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ -1 & 1 & -3 & 0 & 1 & 0 & 9 \end{bmatrix}$$

Row operations: $\mathbf{A}_{(2)} - \frac{3}{2}\mathbf{A}_{(1)}$, $\mathbf{A}_{(3)} - \frac{3}{2}\mathbf{A}_{(1)}$, $\frac{1}{2}\mathbf{A}_{(1)}$

$$\begin{bmatrix} \frac{1}{2} & \frac{3}{2} & 1 & \frac{1}{2} & 0 & 0 & 3 \\ -\frac{1}{2} & -\frac{7}{2} & 0 & -\frac{3}{2} & 1 & 0 & 0 \end{bmatrix}$$

BSF: $\mathbf{x} = [0, 0, 0, 6, 9]^T$, with admissible basis of $\{\mathbf{A}^{(4)}, \mathbf{A}^{(5)}\}$

Pivot in the 3rd column, since $z_3 - c_3 = 3 > 0$

$$\theta = \min \left\{ \frac{|\mathbf{B}^{-1}\mathbf{b}|_i}{x_{i,3}} : x_{i,3} > 0 \right\} = \min \left\{ \frac{6}{2}, \frac{9}{3} \right\} = 3, \text{ either pivoting } x_{1,3} \text{ or } x_{2,3}$$



Example: Degeneracy and Pivot

Example 15

$$\begin{bmatrix} \frac{1}{2} & \frac{3}{2} & 1 & \frac{1}{2} & 0 & 0 & 3 \\ -\frac{1}{2} & -\frac{7}{2} & 0 & -\frac{3}{2} & 1 & 0 & 0 \end{bmatrix}$$

Row operations: $\mathbf{A}_{(2)} + \mathbf{A}_{(1)}$, $\mathbf{A}_{(3)} - \mathbf{A}_{(1)}$, $2\mathbf{A}_{(1)}$

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ 0 & -4 & 1 & -1 & 1 & 0 & 3 \end{bmatrix}$$

New BSF: $\{\mathbf{A}^{(3)}, \mathbf{A}^{(5)}\}$, $\mathbf{x} = [0, 0, 3, 0, 0]^T$ (degenerate)

Pivot in the 1st column, since $z_1 - c_1 = \frac{1}{2} > 0$

$$\theta = \min \left\{ \frac{|\mathbf{B}^{-1}\mathbf{b}|_i}{x_{i,1}} : x_{i,1} > 0 \right\} = \min\{6\} = 6, \text{ pivoting } x_{1,1}$$



Example: Optimal Solution

Example 15

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ 0 & -4 & 1 & -1 & 1 & 0 & 3 \end{bmatrix}$$

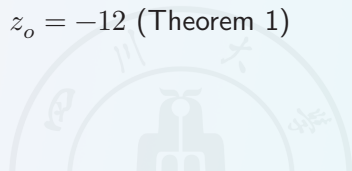
Row operation: $\mathbf{A}_{(2)} - \mathbf{A}_{(1)}$

$$\begin{bmatrix} 1 & 3 & 2 & 1 & 0 & 0 & 6 \\ 0 & -4 & -5 & -1 & -1 & 0 & -12 \end{bmatrix}$$

New BSF: $\{\mathbf{A}^{(1)}, \mathbf{A}^{(5)}\}$, $\mathbf{x} = [6, 0, 0, 0, 3]^T$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x} = [6, 0, 0, 0, 3]^T$ is the optimum, and $z_o = -12$ (Theorem 1)

Original LLP: $\max\{c^T x : x \in F\} = 12$, and $x = [6, 0, 0]^T$



Example: New Problem

Question

$$\min \mathbf{c}^T \mathbf{x}$$

subject to

$$16x_1 + 2x_2 + x_5 + 10x_6 = 4$$

$$-x_1 + x_4 + x_6 = 0$$

$$5x_1 + 4x_2 + x_3 = 8$$

$$x_i \geq 0, \quad \forall i$$

where $\mathbf{c} = [1, -1, 2, -3, 0, 0]^T$



Example: Simplex Steps

Example 16

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ -1 & -1 & -2 & -3 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\downarrow A_{(4)} + (A_{(1)} + A_{(2)} + A_{(3)})$$

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 19 & 5 & 0 & 0 & 0 & 10 & 1 & 12 \end{bmatrix}$$

BSF: $\mathbf{x} = [0, 0, 8, 0, 4, 0]^T$ (degenerate), with admissible basis of $\{\mathbf{A}^{(5)}, \mathbf{A}^{(4)}, \mathbf{A}^{(3)}\}$

Pivot in the 6th column, since $z_6 - c_6 = 10 > 0$

$\theta = \min\{0, \frac{4}{10}\} = 0$, pivoting $x_{2,6}$

Example: Pivot Selection

Example 16

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 19 & 5 & 0 & 0 & 0 & 10 & 1 & 12 \end{bmatrix}$$

$$\downarrow A_{(1)} - 10A_{(2)}, \quad A_{(4)} - 10A_{(2)}$$

$$\begin{bmatrix} 26 & 2 & 0 & -10 & 1 & 0 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 29 & 5 & 0 & -10 & 0 & 0 & 1 & 12 \end{bmatrix}$$

BSF: $\mathbf{x} = [0, 0, 8, 0, 0, 4]^T$ (degenerate), with admissible basis of $\{\mathbf{A}^{(5)}, \mathbf{A}^{(6)}, \mathbf{A}^{(3)}\}$

Pivot in the 2nd column, since $z_2 - c_2 = 5 > 0$

$\theta = \min\{\frac{4}{2}, \frac{8}{4}\} = 2$, pivoting either $x_{1,2}$ or $x_{3,2}$

Example: Further Pivoting

Example 16

$$\begin{bmatrix} 26 & 2 & 0 & -10 & 1 & 0 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 29 & 5 & 0 & -10 & 0 & 0 & 1 & 12 \end{bmatrix}$$

$$\downarrow A_{(3)} - 2A_{(1)}, \quad A_{(4)} - \frac{5}{2}A_{(1)}, \quad \frac{1}{2}A_{(1)}$$

$$\begin{bmatrix} 13 & 1 & 0 & -5 & \frac{1}{2} & 0 & 0 & 2 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -21 & 0 & 1 & 20 & -2 & 0 & 0 & 4 \\ -36 & 0 & 0 & 15 & -\frac{5}{2} & 0 & 1 & 2 \end{bmatrix}$$

BSF: $x = [0, 2, 0, 0, 0, 0]^T$ (degenerate), with admissible basis of $\{\mathbf{A}^{(2)}, \mathbf{A}^{(6)}, \mathbf{A}^{(3)}\}$

Pivot in the 4th column, since $z_4 - c_4 = 15 > 0$

$\theta = \min\{\frac{0}{1}, \frac{0}{20}\} = 0$, pivoting either $x_{2,4}$ or $x_{3,4}$

Example: Optimality Reached

Example 16

$$\begin{bmatrix} 13 & 1 & 0 & -5 & \frac{1}{2} & 0 & 0 & 2 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -21 & 0 & 1 & 20 & -2 & 0 & 0 & 4 \\ -36 & 0 & 0 & 15 & -\frac{5}{2} & 0 & 1 & 2 \end{bmatrix}$$

$$\downarrow A_{(1)} + 5A_{(2)}, \quad A_{(3)} - 20A_{(2)}, \quad A_{(4)} - 15A_{(2)}$$

$$\begin{bmatrix} 8 & 1 & 0 & 0 & \frac{1}{2} & 5 & 0 & 2 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 & -2 & -20 & 0 & 4 \\ -21 & 0 & 0 & 0 & -\frac{5}{2} & -15 & 1 & 2 \end{bmatrix}$$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x} = [0, 2, 0, 0, 0, 0]^T$ is the optimum, and $z_o = 2$ (Theorem 1)

The admissible basis: $\{\mathbf{A}^{(2)}, \mathbf{A}^{(4)}, \mathbf{A}^{(3)}\}$

Example: Multiple Optima

Question: Find several points $\mathbf{x} = [x_1, x_2, x_3, x_4]^T$ that solve the problem

$$\min 3x_1 + 2x_2 + 6x_3 + x_4$$

subject to

$$x_1 + x_4 = 4$$

$$x_1 + x_2 + x_3 = 5$$

$$x_i \geq 0, \forall i$$



Example: Finding BFS

Example 17

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 1 & 1 & 1 & 0 & 0 & 5 \\ -3 & -2 & -6 & -1 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 1 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix}$$

BSF: $\{\mathbf{A}^{(4)}, \mathbf{A}^{(2)}\}, \mathbf{x} = [0, 5, 0, 4]^T$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x}_1 = [0, 5, 0, 4]^T$ is the optimum, and $z_o = 14$ (Theorem 1)

Example: Alternative Optimal Solution

Example 17

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 1 & 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix} \Rightarrow \mathbf{A}_{(2)} - \mathbf{A}_{(1)} \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 4 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix}$$

Consider $z_1 - c_1 = 0$, $\theta = \min\{\frac{4}{1}, \frac{5}{1}\} = 4$, pivoting $x_{1,1}$

BSF: $\{\mathbf{A}^{(1)}, \mathbf{A}^{(2)}\}$, $\mathbf{x} = [4, 1, 0, 0]^T$

Since $z_j - c_j \leq 0, \forall j$, then $\mathbf{x}_2 = [4, 1, 0, 0]^T$ is also the optimum, and $z_o = 14$ (Theorem 1)

Useful Points in Simplex

Useful points when using the primal simplex procedure:

- 1 z_0 should never increase from one tableau to the next
- 2 $\mathbf{B}^{-1}\mathbf{b}$ should never contain negative components
- 3 For each $\mathbf{A}^{(j)}$ in the admissible basis, $z_j - c_j = 0$



Artificial Variables Introduction

To solve LLP with the primal simplex procedure, an initial BFS of F is required before setting up the first tableau.

This is possible after adding slack variables, $\mathbf{A} = [a_{i,j}]$ contains $\mathbf{I}_m$

Question: What if $\mathbf{A} = [a_{i,j}]$ does NOT contain $\mathbf{I}_m$?



Artificial Variables Formulation

Question:

$$\min \quad c_1x_1 + \cdots + c_nx_n + Mx_{n+1} + \cdots + Mx_{n+m}$$

subject to

$$a_{1,1}x_1 + \cdots + a_{1,n}x_n + x_{n+1} = b_1$$

$$a_{2,1}x_1 + \cdots + a_{2,n}x_n + x_{n+2} = b_2$$

$$\vdots$$

$$a_{m,1}x_1 + \cdots + a_{m,n}x_n + x_{n+m} = b_m$$

$$x_1 \geq 0, \dots, x_n \geq 0, x_{n+1} \geq 0, \dots, x_{n+m} \geq 0$$

M : an unspecified large positive number

$x_{n+1}, \dots, x_{n+m}$: artificial variables



Objective with Artificial Variables

- The objective function:

$$f(\mathbf{y}) = c_1 y_1 + \cdots + c_n y_n$$

$$\bar{f}(\bar{\mathbf{y}}) = c_1 y_1 + \cdots + c_n y_n + M y_{n+1} + \cdots + M y_{n+m}$$

- The set of feasible solutions:

$$F = \{\mathbf{y} \in \mathbf{R}_{n \times 1} : \mathbf{A}\mathbf{y} = \mathbf{b}, \mathbf{y} \geq 0\}$$

$$\bar{F} = \{\bar{\mathbf{y}} \in \mathbf{R}_{(n+m) \times 1} : [\mathbf{A}, \mathbf{I}_m]\bar{\mathbf{y}} = \mathbf{b}, \bar{\mathbf{y}} \geq 0\}$$

- Specific solution:

$$[\quad]$$



Mapping between F and $\bar{F}$

$p : F \rightarrow \bar{F}$, i.e., $p(\mathbf{x}) = \bar{\mathbf{x}}$

$p(F) = \{\bar{\mathbf{x}} = p(\mathbf{x}) : \mathbf{x} \in F\}$ is a subset of $\bar{F}$

$\{f(\mathbf{x}) : \mathbf{x} \in F\} = \{\bar{f}(\bar{\mathbf{x}}) : \bar{\mathbf{x}} \in p(F)\}$

$\min\{\bar{f}(\bar{\mathbf{y}}) : \bar{\mathbf{y}} \in \bar{F}\} \leq \min\{f(\mathbf{y}) : \mathbf{y} \in F\}$



Initial Matrix with Artificials

- Objective function: $c_1x_1 + \dots + c_nx_n + Mx_{n+1} + \dots + Mx_{n+m}$
- Initial matrix:

$$\begin{bmatrix} \mathbf{A} & \mathbf{I}_m & \mathbf{0} & \mathbf{b} \\ -\mathbf{c}^T & \mathbf{0}^T & 1 & 0 \\ \mathbf{0}^T & -\mathbf{1}^T & 0 & 0 \end{bmatrix}$$

- The $(m+1)$ -th row $\rightarrow c_1x_1 + \dots + c_nx_n$
- The $(m+2)$ -th row $\rightarrow Mx_{n+1} + \dots + Mx_{n+m}$



First Simplex Tableau

The 1st simplex tableau: row operations on the last row.

$$\left[\begin{array}{c} M \end{array} \right]$$

Reduced Costs with M

- In the tableau:

$$[\quad M \quad]$$

- To use simplex procedure, force each

$$z_j - c_j = a_j M + d_j \leq 0, \quad j = 1, \dots, n + m$$



Simplex with Artificial Variables

For $M \gg 0$ unspecified large:

to force $a_j \leq 0, \forall j$

to force $d_j \leq 0, \forall j$ if $a_j = 0$

Simple procedure involving artificial variables:

- 1 On the last row: force all $a_j^* \leq 0, j = 1, \dots, m + n$ (by iteratively selecting the column with the largest number on the last row and pivoting)
- 2 On the last-but-one row: select $\mathbf{A}^{(j)}$ with the largest positive entry over a 0 in the last row
- 3 Perform simplex procedure



Infeasibility Detection

Suppose a tableau with optimal solution:

If $a_0^* > 0$

The admissible basis contains columns that correspond to artificial variables

$F = \emptyset$: the problem is not feasible, because

$$a_0^*M + d_0^* = \min\{\bar{f}(\bar{\mathbf{y}}) : \bar{\mathbf{y}} \in \bar{F}\} \leq \min\{f(\mathbf{y}) : \mathbf{y} \in F\}$$



Optimal Solution with Artificial

Suppose a tableau with optimal solution:

If $a_0^* = 0$

The admissible basis contains columns that correspond to artificial variables

$\mathbf{x}$ is optimal solution to the original problem:

$$f(\mathbf{x}) = \bar{f}(\bar{\mathbf{x}}) = \min \{ \bar{f}(\bar{\mathbf{y}}) : \bar{\mathbf{y}} \in \bar{F} \} \leq \min \{ f(\mathbf{y}) : \mathbf{y} \in F \}$$



Example: Artificial Variables

Question:

$$\min 2x_1 + x_2 + 3x_3$$

subject to

$$x_1 - 2x_3 = 2$$

$$x_1 + x_3 = 1$$

$$x_2 + x_3 = 1$$

$$\forall x_i \geq 0$$

The coefficient matrix contains $[0, 0, 1]^T$ of $\mathbf{I}_3$. Only need 2 artificial variables.

Example: Transformation

- Transformation:

$$\min 2x_1 + x_2 + 3x_3 + Mx_4 + Mx_5$$

subject to

$$x_1 - 2x_3 + x_4 = 2$$

$$x_1 + x_3 + x_5 = 1$$

$$x_2 + x_3 = 1$$

$$\forall x_i \geq 0$$



Example: Initial Tableau

Initial matrix & the 1st simplex tableau

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ -2 & -1 & -3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(4)} + \mathbf{A}_{(3)}, \quad \mathbf{A}_{(5)} + (\mathbf{A}_{(1)} + \mathbf{A}_{(2)})$$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ -2 & -1 & -3 & 0 & 0 & 1 & 0 & 0 \\ -2 & -1 & 2 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Example: Infeasibility

$$[\quad \quad \quad x^* = [0, 0, 1, 2, 0]^T]$$

Example: Another Artificial Problem

Question:

$$\min 2x_1 + x_2 + x_3 + 3x_4$$

subject to

$$-\frac{1}{3}x_1 + x_3 + x_4 = 1$$

$$x_2 + x_3 = 1$$

$$-x_1 + x_2 + x_3 = 1$$

$$\forall x_i \geq 0$$

Artificial variables: x_5, x_6



Example: Transformation

- Transformation:

$$\min 2x_1 + x_2 + x_3 + 3x_4 + Mx_5 + Mx_6$$

subject to

$$-\frac{1}{3}x_1 + x_3 + x_4 = 1$$

$$x_2 + x_3 + x_5 = 1$$

$$-x_1 + x_2 + x_3 + x_6 = 1$$

$$\forall x_i \geq 0$$

Example: Initial Tableau

Example 19 Admissible basis: $\mathbf{A}^{(4)}, \mathbf{A}^{(5)}, \mathbf{A}^{(6)}$

Initial matrix & the 1st simplex tableau:

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ -2 & -1 & -1 & -3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(4)} + 3\mathbf{A}_{(1)}, \quad \mathbf{A}_{(5)} + (\mathbf{A}_{(2)} + \mathbf{A}_{(3)})$$

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ -3 & -1 & 2 & 0 & 0 & 0 & 1 & 0 & 3 \\ -1 & 2 & 2 & 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$



Example: Pivot

Example 19 the 1st simplex tableau & the 2nd simplex tableau:

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ -3 & -1 & 2 & 0 & 0 & 0 & 1 & 0 & 3 \\ -1 & 2 & 2 & 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(3)} - \mathbf{A}_{(2)}, \mathbf{A}_{(4)} + \mathbf{A}_{(2)}, \mathbf{A}_{(5)} - 2\mathbf{A}_{(2)}$$

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ -3 & 0 & 3 & 0 & 1 & 0 & 1 & 0 & 4 \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Example: Optimal Solution

Example 19 the 2nd simplex tableau & the 3rd simplex tableau ($a_0^* = 0$):

$$\begin{bmatrix} -\frac{1}{3} & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ -3 & 0 & 3 & 0 & 1 & 0 & 1 & 0 & 4 \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(1)} - \mathbf{A}_{(2)}, \quad \mathbf{A}_{(4)} - 3\mathbf{A}_{(2)}$$
$$\begin{bmatrix} -\frac{1}{3} & -1 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ -3 & -3 & 0 & 0 & -2 & 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Basis contains column(s) w.r.t artificial variable(s)

Example: Artificial Variable

Question: $\min 4x_1 + x_2 + x_3 + x_4$
subject to

$$x_1 + x_4 = 4$$

$$x_1 + x_3 = 5$$

$$x_1 + x_2 + x_4 = 9$$

$$\forall x_i \geq 0$$

Artificial variables: x_5



Example: Transformation

- Transformation:

$$\min 4x_1 + x_2 + x_3 + x_4 + Mx_5$$

subject to

$$x_1 + x_4 + x_5 = 4$$

$$x_1 + x_3 = 5$$

$$x_1 + x_2 + x_4 = 9$$

$$\forall x_i \geq 0$$



Example: Initial Tableau

- Admissible basis: $\mathbf{A}^{(3)}, \mathbf{A}^{(3)}, \mathbf{A}^{(2)}$
- Initial matrix & the 1st simplex tableau:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 9 \\ -4 & -1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(5)} + \mathbf{A}_{(1)}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 9 \\ -4 & -1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Example: Optimal Solution

Example 20 the 1st simplex tableau & the 2nd simplex tableau ($a_0^* = 0$):

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 9 \\ -4 & -1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(5)} - \mathbf{A}_{(1)}, \quad \mathbf{A}_{(3)} - \mathbf{A}_{(1)}$$
$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 5 \\ -2 & 0 & 0 & 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Basis does not contain column w.r.t artificial variable

Example: Mixed Constraints

Question:

$$\min -x_1 - x_2 - 3x_3$$

subject to

$$x_1 + x_2 + x_4 = 2$$

$$x_2 + x_3 - x_4 = 1$$

$$x_1 + x_2 + x_4 \leq 5$$

Slack variables: x_4, x_5

Artificial variables: x_6, x_7



Example: Transformation

- Transformation:

$$\min -x_1 - x_2 - 3x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

subject to

$$x_1 + x_2 + x_6 = 2$$

$$x_2 + x_3 - x_4 + x_7 = 1$$

$$x_1 + x_2 + x_4 + x_5 = 5$$

$$\forall x_i \geq 0$$



Example: Initial Tableau

Example 21 Admissible basis: $\mathbf{A}^{(6)}, \mathbf{A}^{(7)}, \mathbf{A}^{(5)}$

Initial matrix & the 1st simplex tableau:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 1 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(5)} + (\mathbf{A}_{(1)} + \mathbf{A}_{(2)})$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 1 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Example: Pivot

the 1st simplex tableau & the 2nd simplex tableau:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 1 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(1)} - \mathbf{A}_{(2)}, \quad \mathbf{A}_{(4)} - \mathbf{A}_{(2)}, \quad \mathbf{A}_{(5)} - 2\mathbf{A}_{(2)}$$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \end{bmatrix}$$

Example: Further Pivoting

the 2nd simplex tableau & the 3rd simplex tableau ($a_0^* = 0$):

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 5 \\ \hline 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 & 0 & 0 & -2 & 0 & 1 & 1 \end{bmatrix}$$

Row operations:

$$\mathbf{A}_{(2)} + \mathbf{A}_{(1)}, \quad \mathbf{A}_{(4)} - \mathbf{A}_{(1)}, \quad \mathbf{A}_{(5)} - \mathbf{A}_{(1)}$$

[]



Example: Optimal Solution

[151 /]

Dual LP Problem

Dual Linear Programming Problems

Primal problem:

$$\min \quad \mathbf{c}^T \mathbf{x}$$

subject to

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ \mathbf{x} &\geq \mathbf{0} \end{aligned}$$

Dual problem:

$$\max \quad \mathbf{b}^T \mathbf{w}$$

subject to

$$\mathbf{A}^T \mathbf{w} \leq \mathbf{c}$$



Weak and Strong Duality

Weak Duality Theorem

If $\mathbf{x}$ and $\mathbf{w}$ are feasible solutions of the primal and dual problems, then $\mathbf{b}^T \mathbf{w} \leq \mathbf{c}^T \mathbf{x}$

Strong Duality Theorem

If $\bar{\mathbf{x}}$ and $\bar{\mathbf{w}}$ are feasible solutions of the primal and dual problems, and $\mathbf{b}^T \bar{\mathbf{w}} = \mathbf{c}^T \bar{\mathbf{x}}$, then

- $\bar{\mathbf{x}}$ and $\bar{\mathbf{w}}$ solve their respective problems
- The optimal values of the primal and the dual objective functions are equal:

$$\begin{aligned}\mathbf{c}^T \bar{\mathbf{x}} &= \min \mathbf{c}^T \mathbf{x} \\ &= \max \mathbf{b}^T \mathbf{w} \\ &= \mathbf{b}^T \bar{\mathbf{w}}\end{aligned}$$



Fundamental Duality Theorem

Fundamental Duality Theorem

- If either the primal/dual problem has a finite optimal solution, then the dual/primal problem has a finite optimal solution as well

Moreover,

$$\min \{ \mathbf{c}^T \mathbf{x} : \mathbf{A} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \} = \max \{ \mathbf{b}^T \mathbf{w} : \mathbf{A}^T \mathbf{w} \leq \mathbf{c} \}$$



Proof of Duality ($\mathbf{x} \rightarrow \mathbf{w}$)

Proof ($\mathbf{x} \rightarrow \mathbf{w}$):

- Assume the BFS $x_B = \mathbf{B}^{-1}\mathbf{b}$ and $x_D = \mathbf{0}^T$, the final simplex tableau is

$$\left[\begin{array}{cc|c} \mathbf{I}_m & \mathbf{B}^{-1}\mathbf{D} & 0 \\ \mathbf{B}^{-1}\mathbf{b} & & \\ \mathbf{0}^T & \mathbf{c}_B^T \mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T & 1 \\ \mathbf{c}_B^T \mathbf{B}^{-1}\mathbf{b} & & \end{array} \right]$$

$$\mathbf{c}_B^T \mathbf{B}^{-1}\mathbf{D} - \mathbf{c}_D^T \leq \mathbf{0}^T, z_o = \mathbf{c}_B^T \mathbf{B}^{-1}\mathbf{b}, [\mathbf{I}_m \mathbf{B}^{-1}\mathbf{D}] = \mathbf{B}^{-1}\mathbf{A}$$

- Define $\mathbf{w} = (\mathbf{c}_B^T \mathbf{B}^{-1})^T \in \mathbf{R}_{n \times 1}$

$$\mathbf{A}^T \mathbf{w} = \mathbf{A}^T (\mathbf{c}_B^T \mathbf{B}^{-1})^T = (\mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{A})^T = (\mathbf{c}_B^T [\mathbf{I}_m \mathbf{B}^{-1} \mathbf{D}])^T = [\mathbf{c}_B^T \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{D}]^T \leq [\mathbf{c}_B^T \mathbf{c}_D^T]^T = \mathbf{c}$$

- Thus, $\mathbf{w}$ is a feasible solution of the dual problem

$$\mathbf{b}^T \mathbf{w} = \mathbf{b}^T (\mathbf{c}_B^T \mathbf{B}^{-1})^T = (\mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b})^T = z_o^T = z_o$$

Proof of Duality ($w \rightarrow x$)

Proof ($w \rightarrow x$):

Denote $w = w_1 - w_2$, and slack variables $w_3 \geq 0$

Reformulation of the dual problem:

$$-\min \quad -b^T w_1 + b^T w_2 + 0^T w_3$$

subject to

$$A^T w_1 - A^T w_2 + I w_3 = c^T, \quad w_1 \geq 0, w_2 \geq 0, w_3 \geq 0$$

$\Downarrow$

$$-A^T w_1 + A^T w_2 - I w_3 = -c^T, \quad w_1 \geq 0, w_2 \geq 0, w_3 \geq 0$$

Dual of Dual

Proof ($\mathbf{w} \rightarrow \mathbf{x}$):

The dual problem of the above:

$$\min \quad [-\mathbf{b} \quad \mathbf{b} \quad \mathbf{0}] \mathbf{x}$$

subject to

$$[-\mathbf{A}^T \quad \mathbf{A}^T \quad -\mathbf{I}] \mathbf{x} \leq [-\mathbf{c}]$$

$$\mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$



Example: Primal-Dual

Question:

$$\min \quad x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

subject to

$$16x_1 + 2x_2 + x_5 + 10x_6 = 4$$

$$-x_1 + x_4 + x_6 = 0$$

$$5x_1 + 4x_2 + x_3 = 8$$

$$\forall x_i \geq 0$$



Example: Dual Formulation

- The dual problem:

$$\begin{array}{ll}\max & 4w_1 + 0w_2 + 8w_3 \\ \text{subject to} & 16w_1 - w_2 + 5w_3 \leq 1 \\ & 2w_1 + 4w_3 \leq 1 \\ & w_3 \leq 1 \\ & w_2 \leq 1 \\ & w_1 \leq 1 \\ & 10w_1 + w_2 \leq 1\end{array}$$

Example: Solving Dual

- The initial simplex tableau:

$$\begin{bmatrix} 16 & 2 & 0 & 0 & 1 & 10 & 0 & 4 \\ -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 5 & 4 & 1 & 0 & 0 & 0 & 0 & 8 \\ 19 & 5 & 0 & 0 & 0 & 10 & 1 & 12 \end{bmatrix}$$

- The final tableau:

$$\begin{bmatrix} 8 & 1 & 0 & 0 & 12 & 5 & 0 & 2 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ -27 & 0 & 1 & 0 & -2 & -20 & 0 & 0 \\ -21 & 0 & 0 & 0 & -5/2 & -15 & 1 & 2 \end{bmatrix}$$

- $w^T = c_B^T B^{-1} = c_B^T B^{-1} D - c_D^T + c_D^T$, if $D = I$
 $w_1 = -5/2 + 1 = -3/2$, $w_2 = 0 + 1 = 1$, $w_3 = 0 + 1 = 1$
- The optimal value of the cost function is 2.

